SIEWS+ 5.0 — Model Performance Documentation

AI-Based Detection Pipeline: Pretrained & Fine-tuned Models

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Contents

1 Overview

SIEWS+ 5.0 menggunakan multi-stage detection pipeline yang terdiri dari 5 tahap utama ditambah modul vehicle detection. Pipeline ini menggabungkan model **pretrained** (dari Ultralytics dan HuggingFace) dengan model **fine-tuned** pada dataset khusus industri migas.

Table 1: Ringkasan Pipeline Detection SIEWS+ 5.0

Stage	Task	Model	Type	mAP@50
1 (Live)	Person Detection	YOLO26n	Pretrained	∼56%
1 (Photo)	Person Detection	YOLOv8l	Pretrained	∼69%
2	PPE Detection	YOLOv8s (custom)	Fine-tuned	—
3	Open Hole	YOLOv8n (custom)	Fine-tuned	—
4	Road Damage	YOLOv8s (custom)	Fine-tuned	—
5	Fire & Smoke	YOLOv26s	Pretrained	**94.9%**
—	Vehicle	YOLOv8s COCO	Pretrained	∼62%

2 Pretrained Models

2.1 Stage 1 — Person Detection (Live): YOLO26n

Table 2: YOLO26n / YOLO11n Performance on COCO val2017

Metric	Value
Model File	`yolo26n.pt`
Architecture	YOLO11 Nano
Dataset	MS COCO (80 classes)
Input Size	640×640
mAP@50-95	39.5
Parameters	2.6M
FLOPs	6.5B
Speed (CPU ONNX)	56.1 ms
Speed (T4 TensorRT)	1.5 ms

Source: Ultralytics YOLO11 Documentation
`https://docs.ultralytics.com/models/yolo11/`
Download: `https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt`

Dipilih untuk live camera karena kecepatan inferensi sangat tinggi (1.5ms GPU) yang memungkinkan 30+ FPS real-time processing.

2.2 Stage 1 — Person Detection (Photo): YOLOv8l

Table 3: YOLOv8l Performance on COCO val2017

Metric	Value
Model File	`yolov8l_coco.pt`
Architecture	YOLOv8 Large
Dataset	MS COCO (80 classes)
Input Size	640×640
mAP@50-95	52.9
Parameters	43.7M
FLOPs	165.2B
Speed (CPU ONNX)	375.2 ms
Speed (T4 TensorRT)	6.4 ms

Source: Ultralytics YOLOv8 Documentation
`https://docs.ultralytics.com/models/yolov8/`
Download: `https://github.com/ultralytics/assets/releases/download/v8.2.0/yolov8l.pt`

Digunakan untuk analisis foto statis dimana akurasi lebih penting dari kecepatan. Model terbesar di keluarga YOLOv8 dengan mAP tertinggi.

2.3 Stage 5 — Fire & Smoke Detection: YOLOv26s

Table 4: YOLOv26s Fire/Smoke Detection Performance

Metric	Value
Model File	`fire_smoke_yolov26s_94mAP.pt`
Architecture	YOLOv26 Small
Dataset	Custom Fire/Smoke
Input Size	640×640
mAP@50	**94.9%**
Classes	fire, smoke, other
Parameters	∼9M

Source: HuggingFace — SalahALHaismawi/yolov26-fire-detection
`https://huggingface.co/SalahALHaismawi/yolov26-fire-detection`

Model pretrained khusus deteksi api dan asap dengan akurasi sangat tinggi (94.9% mAP@50). Berbasis arsitektur YOLOv26 terbaru.

2.4 Vehicle Detection: YOLOv8s COCO

Table 5: YOLOv8s Performance on COCO val2017

Metric	Value
Model File	`yolov8s_coco.pt`
Architecture	YOLOv8 Small
Dataset	MS COCO (filter: car, bus, truck, motorcycle)
Input Size	640×640
mAP@50-95	44.9
Parameters	11.2M
FLOPs	28.6B
Speed (CPU ONNX)	98.9 ms
Speed (T4 TensorRT)	2.3 ms

Source: Ultralytics YOLOv8 Documentation
`https://docs.ultralytics.com/models/yolov8/`
Download: `https://github.com/ultralytics/assets/releases/download/v8.2.0/yolov8s.pt`

Menggunakan COCO pretrained dan filter hanya kelas kendaraan (car=2, motorcycle=3, bus=5, truck=7). Keseimbangan baik antara speed dan accuracy.

3 Fine-tuned Models

Model berikut di-train pada dataset khusus menggunakan transfer learning dari pretrained weights.

3.1 Stage 2 — PPE Detection

Table 6: PPE Detection Model Configuration

Parameter	Value
Model File	`best_stage2_labeled_safety.pt`
Base Model	YOLOv8s
Classes	helmet, vest, belt, unknown
Training Script	`training/train_stage2.py`
mAP	Requires local evaluation

3.2 Stage 3 — Environment Hazard (Open Hole)

Table 7: Open Hole Detection Model Configuration

Parameter	Value
Model File	`best_stage3_openhole.pt`
Base Model	YOLOv8n
Classes	open-hole
Training Script	`training/train_stage3.py`
mAP	Requires local evaluation

3.3 Stage 4 — Road Damage (Pothole)

Table 8: Road Damage Detection Model Configuration

Parameter	Value
Model File	`best_jalan_berlubang.pt`
Base Model	YOLOv8s
Classes	pothole / lubang jalan
Training Script	`training/train_stage4.py`
mAP	Requires local evaluation

Untuk mendapatkan mAP model fine-tuned, jalankan:

```
yolo val detect model=path/to/model.pt data=path/to/dataset.yaml
```

4 Performance Comparison Charts

4.1 YOLO Family — mAP@50-95 vs Parameters

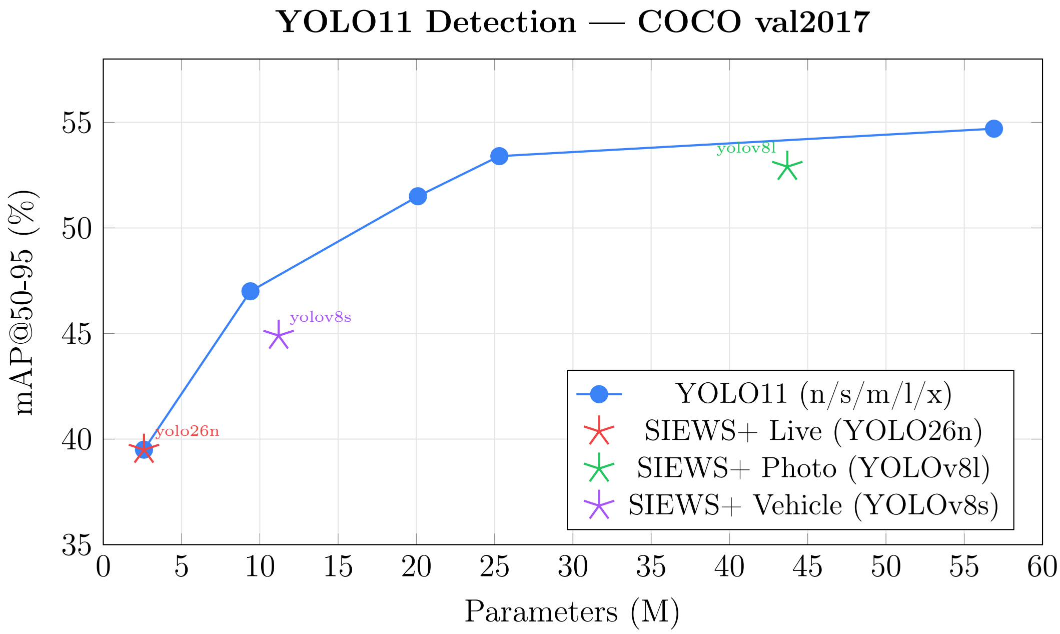

Figure 1: Perbandingan mAP vs jumlah parameter. Model yang digunakan SIEWS+ ditandai bintang.

4.2 YOLO Family — mAP@50-95 vs Inference Speed

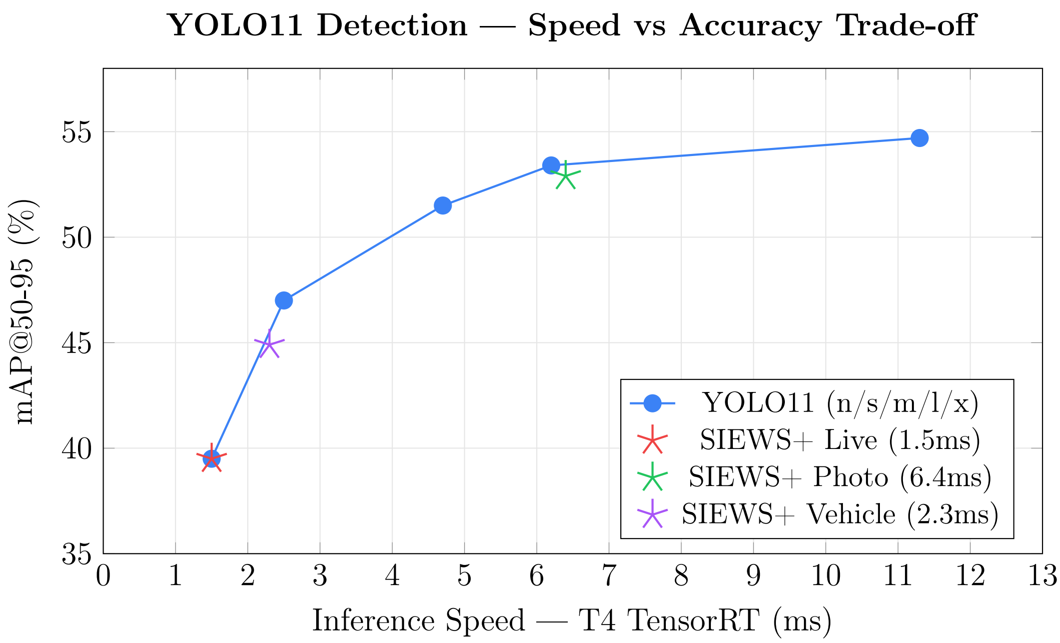

Figure 2: Trade-off antara kecepatan inferensi dan akurasi. SIEWS+ Live menggunakan model tercepat untuk real-time 30fps.

4.3 SIEWS+ Pipeline — mAP@50 per Stage

Figure 3: mAP@50 untuk setiap model pretrained yang digunakan dalam pipeline SIEWS+.

4.4 Full YOLO Family Benchmark (Reference)

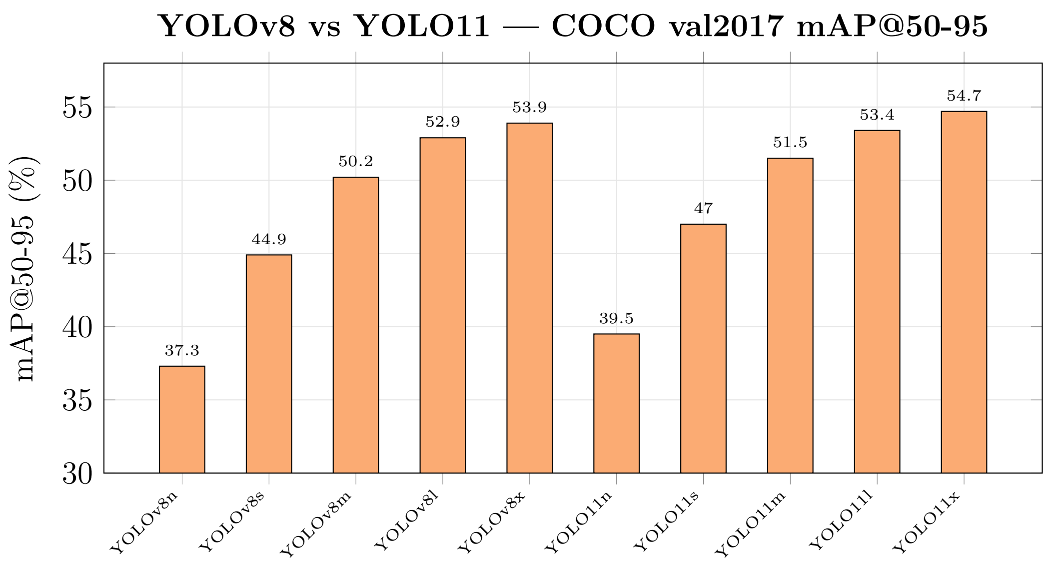

Figure 4: Perbandingan lengkap YOLOv8 dan YOLO11 pada COCO val2017. YOLO11 konsisten lebih baik di setiap ukuran model.

5 References

1. Ultralytics YOLOv8 Documentation — `https://docs.ultralytics.com/models/yolov8/`
2. Ultralytics YOLO11 Documentation — `https://docs.ultralytics.com/models/yolo11/`

3. YOLO11 HuggingFace Model Card — `https://huggingface.co/Ultralytics/YOLO11`

4. SalahALHaismawi, YOLOv26 Fire Detection (94.9% mAP@50) — `https://huggingface.co/SalahALHaismawi/yolov26-fire-detection`

5. MS COCO Dataset — `https://cocodataset.org/`

6. Ultralytics GitHub Repository — `https://github.com/ultralytics/ultralytics`

7. COCO Detection Benchmark — `https://docs.ultralytics.com/datasets/detect/coco/`

Note: mAP values for pretrained models are sourced from official Ultralytics documentation and HuggingFace model cards. Fine-tuned model performance requires local evaluation on respective validation datasets.